

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Error Analysis Task

COURSE NAME: Cloud Computing and Big Data Analytics

COURSE CODE: CSA15

COURSE OUTCOME COVERED

CO5: Design big data processing systems using the Hadoop ecosystem including HDFS, MapReduce, and YARN.
(BL5)

Dave's Taxonomy: Articulation (Level 4)

Question Count: 5

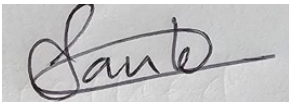
Total Marks: 25

CODE OF CONDUCT

I SANDEEP K / 192521253 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



SANDEEP K / 192521253

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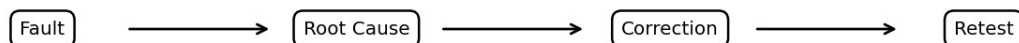
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1. Uneven input splits and idle reducers

Repeated failures indicate workload imbalance or key skew. Some tasks become overloaded while other reducers finish early.

- Create balanced input splits.
- Avoid unsuitable file/split sizes.
- Use a suitable partitioner to reduce skew.
- Use a Combiner where valid.
- Monitor mapper/reducer load and tune reducer count.

Error Analysis Flow

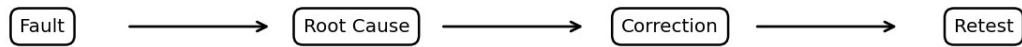


2. HDFS unavailable after one node failure

A single node failure should not normally cause data loss. Weak replication, poor placement or NameNode availability is the likely issue.

- Verify replication and under-replicated blocks.
- Use rack-aware replica placement.
- Deploy NameNode high availability.
- Trigger re-replication after failure.
- Monitor DataNode and NameNode health.

Error Analysis Flow

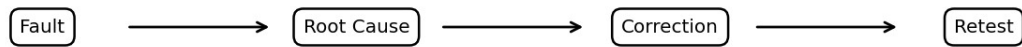


3. Duplicate records from ETL/NiFi

Duplicates commonly occur when retries or repeated flows insert the same record. Ingestion should be traceable and idempotent.

- Use stable unique record identifiers.
- Track NiFi flow/file state.
- Use upsert or merge operations.
- Add target-side deduplication.
- Audit duplicate counts and retries.

Error Analysis Flow

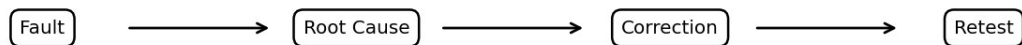


4. YARN resource starvation

Poor queue configuration can allow some jobs to consume most resources. Scheduler policies should enforce controlled sharing.

- Create queues with minimum/maximum resources.
- Use fair or capacity scheduling.
- Set realistic resource requests.
- Use priorities without starving other queues.
- Monitor utilization and tune the policy.

Error Analysis Flow



5. Outdated data after recovery

Stale recovery data indicates that backups do not match the required recovery point. Live replicas alone are not a complete historical backup.

- Use scheduled, versioned backups.
- Define recovery point and recovery time objectives.
- Verify backup timestamps and completeness.
- Keep independent backup copies.
- Perform regular restore tests.

Error Analysis Flow

